Infrastructure Steward Runbook
Author: Ilkin Samadov
Role: System Administrator
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## 1. Purpose & Overview

This runbook documents the build, operation, backup, restore, and monitoring procedures for the Infrastructure Steward server. The server provides internal DNS services, a hardened SSH jump-box, automated configuration backups, and basic monitoring.

## 2. System Build Summary

OS: Ubuntu Server 22.04 LTS
Hostname: infra-steward
Subnet: 192.168.56.0/24
IP Address: 192.168.56.102
Users & Roles
opsadmin: Sudo-enabled operations user
auditor: Non-sudo audit user
Installed Services
SSH, BIND9 (DNS), Fail2Ban, UFW firewall, Chrony, Cron

## 3. Security Configuration

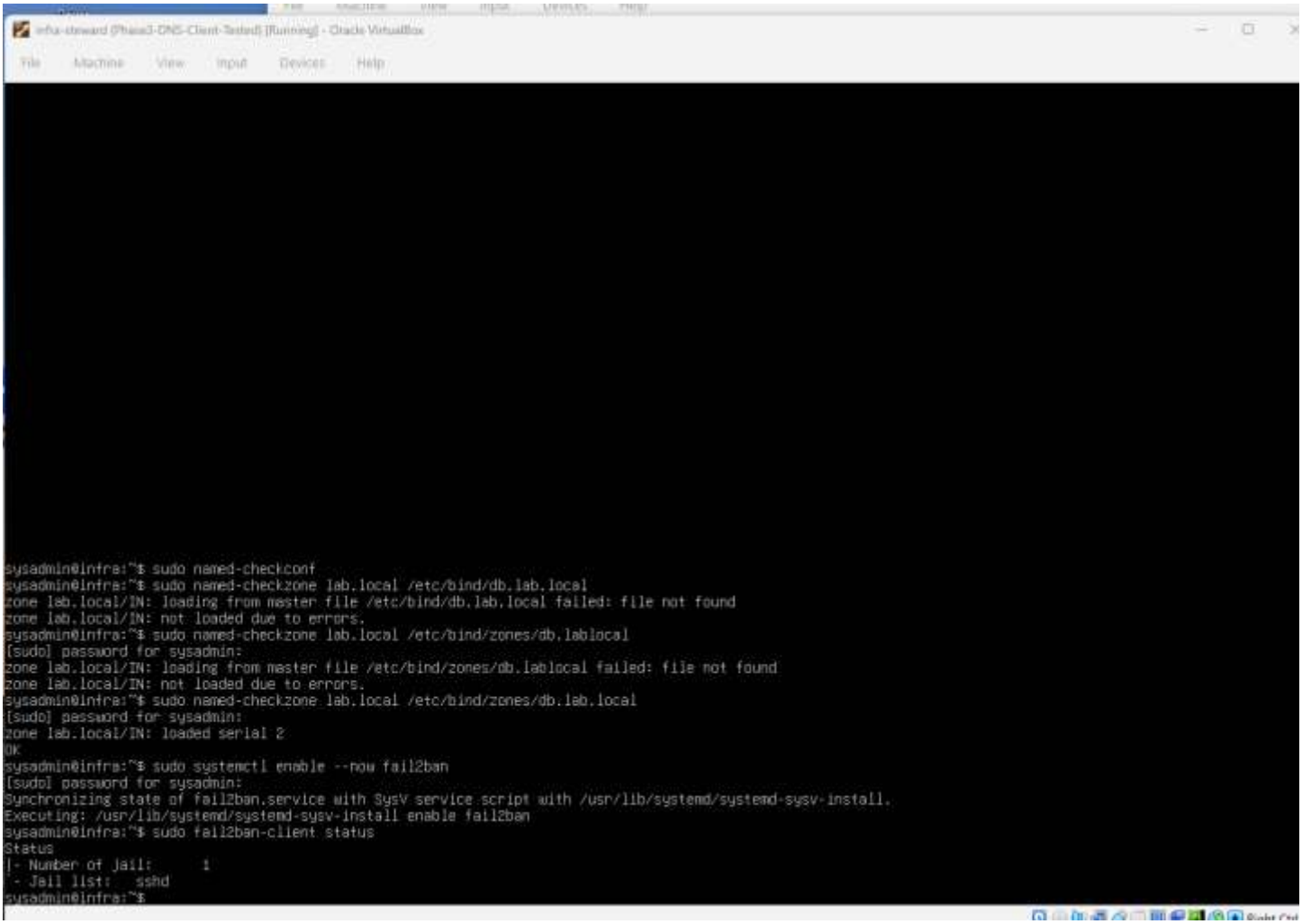
SSH is hardened with root login disabled and limited authentication attempts. Firewall rules allow only SSH and DNS from the internal subnet.

```
sysadmin@infra:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)

sysadmin@infra:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)

sysadmin@infra:~$ sudo systemctl enable --now failban
[sudo] password for sysadmin:
Failed to enable unit: Unit file failban.service does not exist.
sysadmin@infra:~$ sudo update
sudo: update: command not found
sysadmin@infra:~$ sudo apt update
Hit:1 http://us.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:3 http://us.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://us.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 126 kB in 1s (162 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
sysadmin@infra:~$ sudo apt install -y fail2ban
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
fail2ban is already the newest version (1.0.2-3ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
sysadmin@infra:~$ sudo systemctl -y fail2ban
systemctl: invalid option -- 'y'
sysadmin@infra:~$ sudo systemctl enable --now fail2ban
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
sysadmin@infra:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
`- Jail list:   sshd
sysadmin@infra:~$
```

4. Monitoring

A healthcheck script validates DNS, SSH, and disk usage every five minutes and logs results to the system journal.5. Daily Operations Checks

Verify service status, logs, Fail2Ban activity, disk usage, and pending updates daily.




6. Backup & Restore

Daily configuration backups run via cron. Restores are performed from the latest backup to quickly recover DNS configurations.

7. Monitoring

A healthcheck script validates DNS, SSH, and disk usage every five minutes and logs results to the system journal.

8. Troubleshooting

Check bind9 and ssh services, validate configurations, review firewall rules, and confirm client DNS

settings.

```
infra-steward (Phase2-DNS-Working) [Running] - Oracle VirtualBox
File   Machine   View   Input   Devices   Help
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


sysadmin@infra:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:44:6c:1a brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s3
       valid_lft 481sec preferred_lft 481sec
    inet6 fe80::a00:27ff:fe44:6c1a/64 scope link
       valid_lft forever preferred_lft forever
sysadmin@infra:~$ sudo systemctl status bind9 --no-pager
[sudo] password for sysadmin:
● named.service - BIND Domain Name Server
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-01-12 20:47:59 UTC; 8min ago
       Docs: man:named(8)
   Main PID: 836 (named)
     Status: "running"
      Tasks: 8 (limit: 2265)
     Memory: 31.1M (peak: 31.4M)
        CPU: 193ms
     CGroup: /system.slice/named.service
             └─836 /usr/sbin/named -f -u bind

Jan 12 20:47:59 infra named[836]: configuring command channel from '/etc/bind/rndc.key'
Jan 12 20:47:59 infra named[836]: command channel listening on ::1#953
Jan 12 20:47:59 infra named[836]: managed-keys-zone: loaded serial 6
Jan 12 20:47:59 infra named[836]: zone 0.in-addr.arpa/IN: loaded serial 1
Jan 12 20:47:59 infra named[836]: zone 255.in-addr.arpa/IN: loaded serial 1
Jan 12 20:47:59 infra named[836]: zone localhost/IN: loaded serial 2
Jan 12 20:47:59 infra named[836]: zone 127.in-addr.arpa/IN: loaded serial 1
Jan 12 20:47:59 infra named[836]: all zones loaded
Jan 12 20:47:59 infra systemd[1]: Started named.service - BIND Domain Name Server.
Jan 12 20:47:59 infra named[836]: running
sysadmin@infra:~$ sudo ufw allow from 192.168.56.0/24 to any port 53 proto udp
Rule added
sysadmin@infra:~$ sudo ufw allow from 192.168.56.0/24 to any port 53 proto udp
Skipping adding existing rule
sysadmin@infra:~$ sudo ufw reload
Firewall reloaded
sysadmin@infra:~$ _
```

9. Conclusion

The Infrastructure Steward server is secure, monitored, backed up, and fully documented.